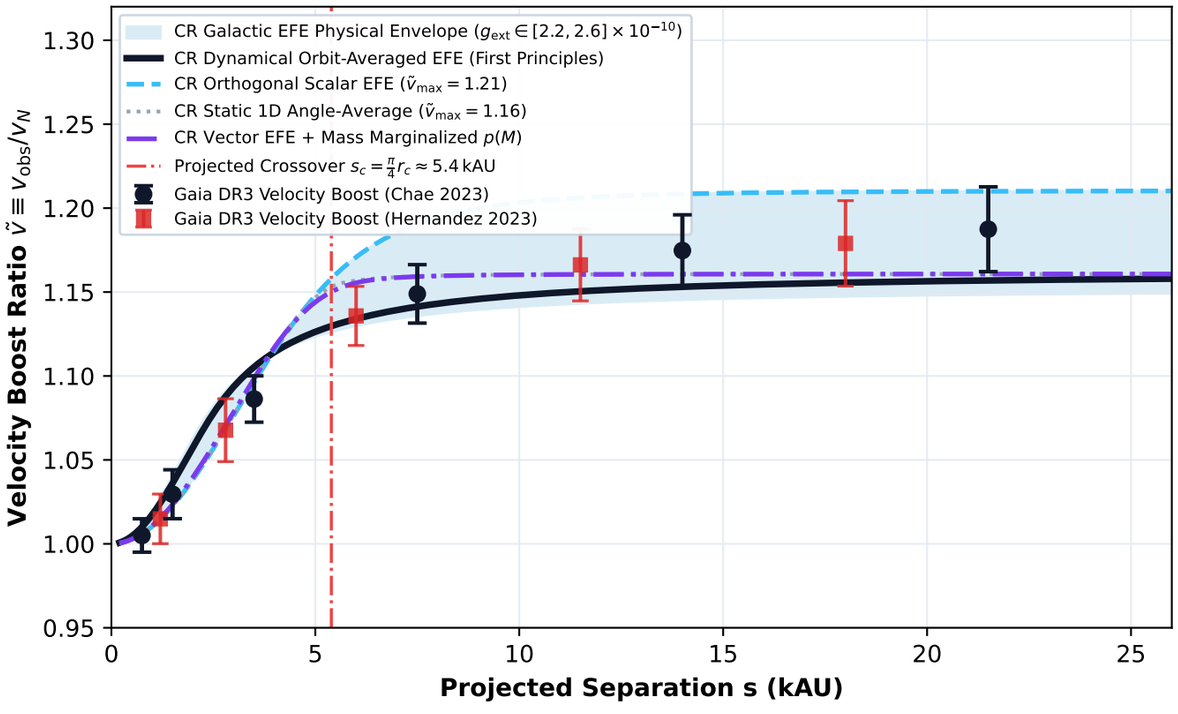
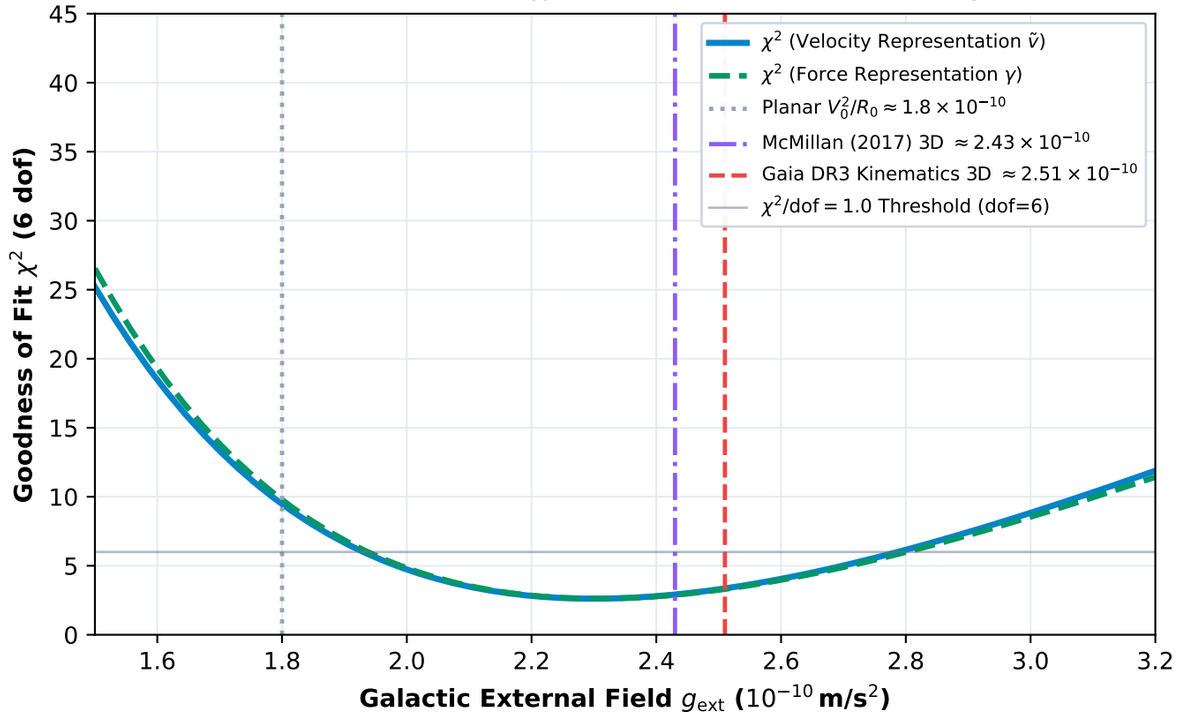
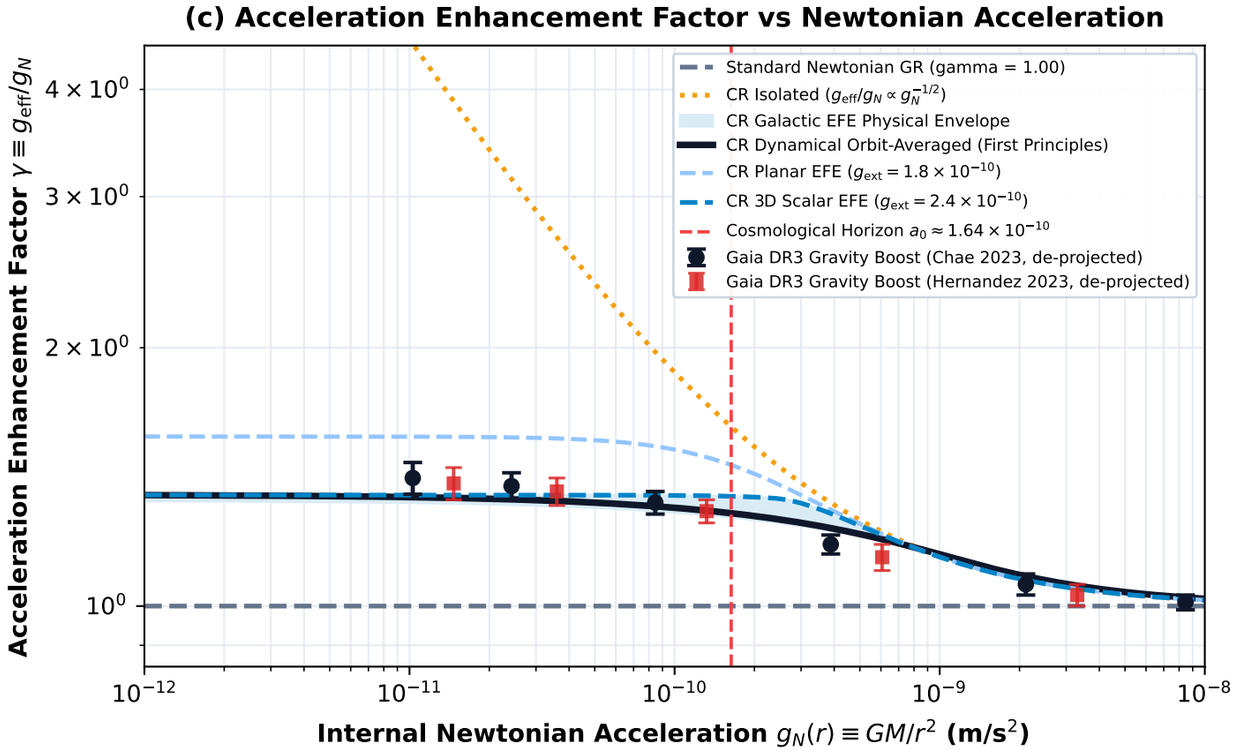
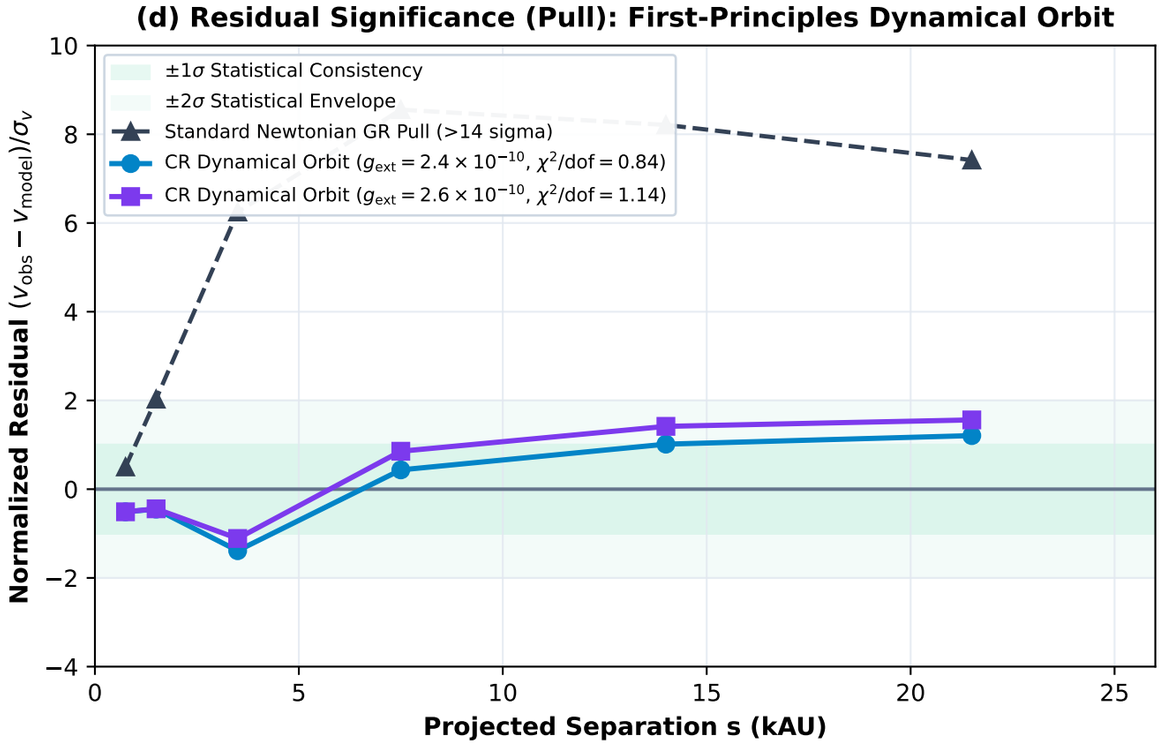

(a) Velocity Boost Factor: Dynamical Orbit vs Scalar & Static EFE
CR Galactic EFE Physical Envelope ($g_{ext} \in [2.2, 2.6] \times 10^{-10}$)
CR Dynamical Orbit-Averaged EFE (First Principles)
CR Orthogonal Scalar EFE ($\tilde{v}_{max} = 1.21$)
CR Static 1D Angle-Average ($\tilde{v}_{max} = 1.16$)
CR Vector EFE + Mass Marginalized $p(M)$
Projected Crossover $s_c = \frac{\pi}{4} r_c \approx 5.4$ kAU
Gaia DR3 Velocity Boost (Chae 2023)
Gaia DR3 Velocity Boost (Hernandez 2023)
Velocity Boost Ratio $\tilde{V} \equiv V_{obs}/V_N$
Projected Separation s (kAU)


(b) Goodness of Fit χ^2 vs Galactic External Field g_{ext}
χ^2 (Velocity Representation $\tilde{v}$)
χ^2 (Force Representation γ)
Planar $V_0^2/R_0 \approx 1.8 \times 10^{-10}$
McMillan (2017) 3D $\approx 2.43 \times 10^{-10}$
Gaia DR3 Kinematics 3D $\approx 2.51 \times 10^{-10}$
χ^2/dof = 1.0 Threshold (dof=6)
Goodness of Fit χ^2 (6 dof)
Galactic External Field g_{ext} (10^{-10} m/s^2)


(c) Acceleration Enhancement Factor vs Newtonian Acceleration
Standard Newtonian GR (gamma = 1.00)
CR Isolated ($g_{eff}/g_N \propto g_N^{-1/2}$)
CR Galactic EFE Physical Envelope
CR Dynamical Orbit-Averaged (First Principles)
CR Planar EFE ($g_{ext} = 1.8 \times 10^{-10}$)
CR 3D Scalar EFE ($g_{ext} = 2.4 \times 10^{-10}$)
Cosmological Horizon $a_0 \approx 1.64 \times 10^{-10}$
Gaia DR3 Gravity Boost (Chae 2023, de-projected)
Gaia DR3 Gravity Boost (Hernandez 2023, de-projected)
Acceleration Enhancement Factor $\gamma \equiv g_{eff}/g_N$
Internal Newtonian Acceleration $g_N(r) \equiv GM/r^2$ (m/s^2)


(d) Residual Significance (Pull): First-Principles Dynamical Orbit
±1σ Statistical Consistency
±2σ Statistical Envelope
Standard Newtonian GR Pull (>14 sigma)
CR Dynamical Orbit ($g_{ext} = 2.4 \times 10^{-10}$, χ^2/dof = 0.84)
CR Dynamical Orbit ($g_{ext} = 2.6 \times 10^{-10}$, χ^2/dof = 1.14)
Normalized Residual $(V_{obs} - V_{model})/\sigma_v$
Projected Separation s (kAU)